Python If-Else: 40 Practice Questions

Student worksheet • Questions only • Beginner to advanced

Instructions: Solve each question using conditional statements. Start with fixed values; later, try taking values using `input()`. Focus on correct conditions and indentation.

A. Simple if Statements

1. Write a program to check whether a number is greater than 10.
2. Write a program to check whether a person is eligible to vote (age must be 18 or above).
3. Write a program to check whether a number is divisible by 5.
4. Write a program to check whether a student has scored at least 35 marks.
5. Write a program to check whether a character is an uppercase letter.

B. if-else Statements

6. Write a program to check whether a number is even or odd.
7. Write a program to check whether a number is positive or non-positive.
8. Write a program to check voting eligibility using if-else.
9. Write a program to check whether a number is divisible by both 3 and 5.
10. Write a program to check whether two numbers are equal or different.
11. Write a program to display Pass if marks are 35 or above; otherwise display Fail.
12. Write a program to check whether a year is a leap year.

C. if-elif-else Statements

13. Write a program to identify whether a number is positive, negative, or zero.
14. Write a program to find the greatest of three numbers.
15. Write a program to display a grade using these ranges: A = 90–100, B = 75–89, C = 60–74, Pass = 35–59, Fail = below 35. Also handle marks outside 0–100 as invalid.
16. Write a program to classify a character as an uppercase letter, lowercase letter, digit, or special character.
17. Write a program to display the weekday for a number from 1 to 7; display Invalid for any other number.
18. Write a program to display the number of days in a month, assuming February has 28 days.
19. Write a program to calculate a discount: 20% for a purchase of ■5,000 or more, 10% for ■2,000 or more, otherwise no discount.
20. Write a program to classify a triangle as equilateral, isosceles, or scalene, assuming the three sides form a valid triangle.

D. Nested if Statements

21. Write a program to check a username first and then check the password only if the username is correct.
22. Write a program to check whether a person is at least 18 years old and then check whether they are registered to vote.
23. Write a program to check whether a student has passed; if passed, check whether the student has scored at least 75 marks for distinction.

24. Write a program to check whether a number is positive; if positive, check whether it is even or odd.

25. Write a program to check whether an applicant meets the minimum age requirement; if so, check whether their salary is sufficient for a loan.

26. Write a login program that checks the username, then password, and finally whether the account is active.

E. Logical Operators (and, or, not)

27. Write a program to check whether a number is between 10 and 50 inclusive.

28. Write a program to check whether a number is divisible by 3 or 7.

29. Write a program to check job eligibility: age must be from 21 to 35 inclusive and the candidate must have the required qualification.

30. Write a program to check whether a character is a vowel using the or operator.

31. Write a program to check whether a number is not between 100 and 200.

32. Write a program to display Pass only if a student passes both theory and practical exams.

33. Write a program to provide free delivery if the order amount is at least ■500 or the customer has premium membership.

F. Advanced Conditional Practice

34. Write a program to find the greatest of four numbers.

35. Write a program to calculate an electricity bill using these slabs: first 100 units at ■2/unit, next 100 units at ■3/unit, and units above 200 at ■5/unit. Apply each rate only to the units in that slab.

36. Write a program to calculate an employee bonus based on years of service: 10 or more years = 15%, 5–9 years = 10%, fewer than 5 years = 5%.

37. Write a program to check whether three angles form a valid triangle (sum must be 180 and each angle must be positive); if valid, classify it as acute, right, or obtuse.

38. Write a program to calculate a purchase discount first, then add 18% tax to the discounted amount. Choose and clearly state your own discount rules.

39. Write a program to validate a time using hours, minutes, and seconds (hours 0–23, minutes 0–59, seconds 0–59).

40. Write an ATM withdrawal program that checks the PIN, then checks whether the balance is sufficient, and finally ensures the withdrawal amount is positive and a multiple of ■100.